Entropically Regularized Martingale Optimal Transport with L_1 Relaxation

Ethan Chen
Georgia Institute of Technology · echen347@gatech.edu

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Project page

Contributions

Setting. Martingale OT requires $\mathbb{E}_\pi[Y|X] = X$ on couplings $\pi \in \Pi(\mu,\nu)$, yielding model-free no-arbitrage pricing bounds. By Strassen's theorem, feasibility requires $\mu \preceq_{cx} \nu$ — generically violated by empirical marginals — and exact martingale couplings form a measure-zero slice of $\Pi(\mu,\nu)$.

1. **Primal-dual formulation.** Replace the hard martingale constraint with an L_1 penalty on the defect $d(\pi)$; dual is leverage-bounded hedging via $-\delta\|h\|_\infty$.
2. **Proximal gradient algorithm.** Sinkhorn inner loop, L_∞-prox outer step; $O(1/k)$ convergence (Prop. 3.1).
3. **Concentration bound.** Empirical martingale violations scale as $O(N^{-1/2})$ (Thm. 3.1), giving a principled lower bound δ_N for the relaxation.

Problem Formulation

Given marginals μ,ν and cost C, the **martingale defect** at x_i is:

$$d_i(\pi) = \mathbb{E}_\pi[Y \mid X = x_i] - x_i = \frac{1}{\mu_i}\sum_j \pi_{ij}(y_j - x_i)$$

The **relaxed MOT** with entropic regularization:

$$\min_{\pi\in\Pi(\mu,\nu)} \langle C,\pi\rangle + \varepsilon H(\pi) \quad \text{s.t.} \quad \|d(\pi)\|_{L_1(\mu)} \le \delta$$

- $\delta = 0$ recovers hard-constrained MOT; $\varepsilon > 0$ gives Sinkhorn-type entropic smoothing
- **Choice of $L_1(\mu)$:** L_2 amplifies pointwise spikes; L_∞ is overly restrictive. L_1 yields a **bounded** dual feasible set via $\|v\|_1 = \sup_{\|h\|_\infty\le 1}\langle h, v\rangle$.

Dual Analysis & Financial Interpretation

Via minimax and Sion's theorem, the dual problem is:

$$\max_{h\in\mathbb{R}^n} \; W_\varepsilon(\tilde{C}(h)) - \delta\|h\|_\infty$$

where $\tilde{C}_{ij}(h) = C_{ij} + h_i(y_j - x_i)$ and W_ε is the entropic OT value.

Financial interpretation: h is a static hedge in the underlying; $-\delta\|h\|_\infty$ imposes an implicit **leverage bound**. As $\delta \to 0$, unbounded hedges are needed — recovering the classical MOT dual. Parallels pricing under proportional transaction costs (Dolinsky–Soner '14).

Algorithm: Proximal Gradient Ascent

The dual decomposes as smooth + non-smooth, amenable to proximal methods:

1. **Gradient step:** $\nabla_h W_\varepsilon = d(\pi^\star(h))$ via **Sinkhorn** (Danskin's thm)
2. **Proximal step:** $\mathrm{prox}_{\tau\|\cdot\|_\infty}(v) = v - \tau\cdot\mathrm{proj}_{\mathcal{B}_1}(v/\tau)$

Convergence (Prop. 3.1): For $|y_j - x_i| \le B$, with step size $\gamma = \varepsilon/B^2$: $J(h^\star) - J(h^{(K)}) \le \frac{B^2\|h^\star\|_2^2}{2\varepsilon K}$

Convergence Verification

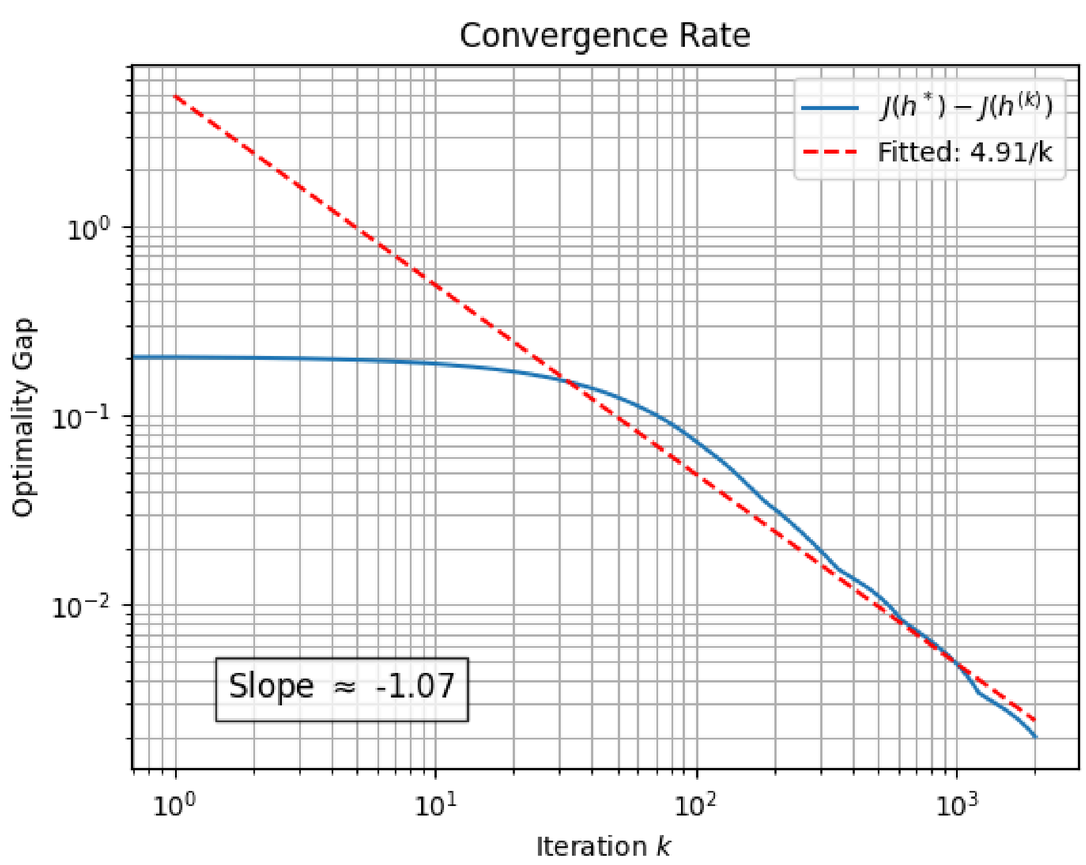

Figure 2. Dual optimality gap vs. iteration. Observed decay matches the $O(1/k)$ rate of Proposition 3.1.

Main Theorem: Principled Relaxation Scale

Question. How small can δ be before statistical noise makes the problem infeasible?

Theorem 3.1 (McDiarmid concentration of empirical defect)

$$\mathbb{P}\Big(\Big|\frac{D_N}{N} - \mathbb{E}|Y-X|\Big| \ge t\Big) \le 2\exp\Big(-\frac{Nt^2}{2B^2}\Big)$$

Inverting at confidence α gives the **principled lower bound**:

$$\delta_N = \sqrt{2}\,B\sqrt{\log(2/\alpha)/N} \;=\; O(N^{-1/2})$$

Proof via bounded-differences (change one Z_i alters D_N/N by $\le 2B/N$).

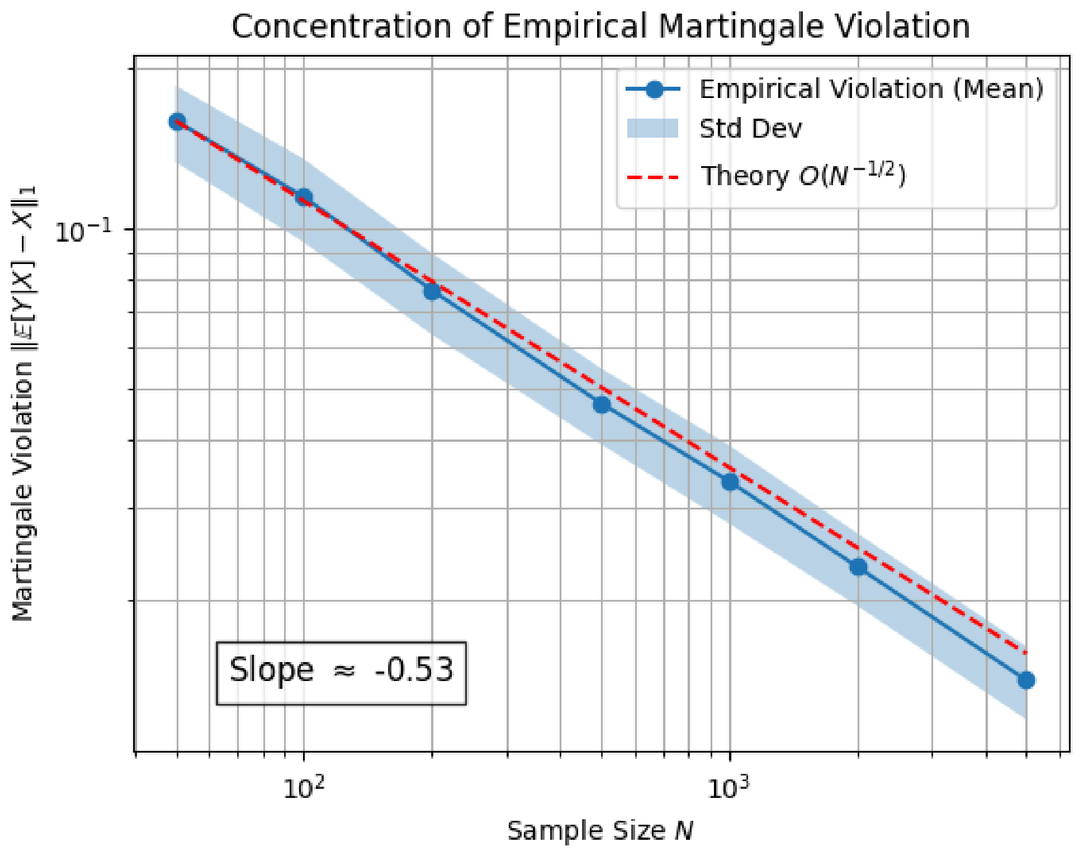

Figure 1. Log-log empirical martingale violation vs. N (50 trials). Observed slope ≈ -0.53 confirms the $O(N^{-1/2})$ rate.

Quantitative Results

Synthetic 1-D Gaussian data: $\mu = \mathcal{N}(0, 0.5^2)$, $\nu = \mathcal{N}(0, 1^2)$, quadratic cost, $\varepsilon = 0.05$.

Method	Transport Cost	Martingale Violation
Unconstrained Sinkhorn	0.2637	0.3746
Relaxed MOT ($\delta = 0.1$)	0.5041	0.1099
Relaxed MOT ($\delta = 0.01$)	0.6573	**0.0269**

Tightening δ from 0.1 to 0.01 reduces violation by $4\times$ with only ~30% cost increase.

Relaxation Trade-off

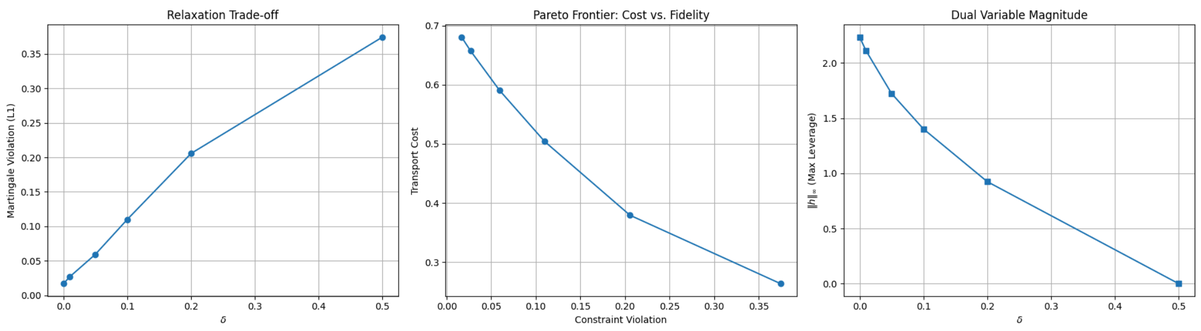

Figure 3. (Left) Martingale violation vs. δ. (Middle) Pareto frontier: transport cost vs. violation. (Right) Dual variable norm $\|h\|_\infty$ vs. δ — divergence as $\delta \to 0$ confirms the leverage-bound interpretation.

Takeaways · Limits · Next

Takeaways. Smooth Pareto frontier between transport cost and martingale fidelity. $\|h\|_\infty \to \infty$ as $\delta \to 0$ confirms leverage-bound duality. Both rates $O(1/k)$, $O(N^{-1/2})$ verified.

Limits. 1D synthetic Gaussians only; no real options data. Thm. 3.1 needs bounded support (excludes log-normals). $\varepsilon = 0.05$ fixed; FISTA unevaluated.

Next. Multi-marginal MOT (non-convex for ≥ 3 marginals). Adaptive δ (Lepski). DRO via Dolinsky–Soner duality. Sub-Gaussian extension of Thm. 3.1.

Project: echen347.github.io/mot
Contact: echen347@gatech.edu